ECE / OPEN RESEARCH PROJECT

Edge AI for Electric-Vehicle Battery Health

Build an embedded diagnostic system that estimates lithium-ion battery state-of-health from current, voltage and temperature signals without uploading raw telemetry.

TEAM	DURATION	FORMAT
Up to 4 students	14-week prototype	Mentored research

Problem

Battery degradation is nonlinear and depends on temperature, load history and cell variation. Cloud-only diagnostics introduce latency, connectivity and privacy constraints for fleet-scale monitoring.

Method

The team will curate cycling traces, engineer impedance and temporal features, compare compact sequence models, quantize the best model for an ARM-class device and validate estimates against controlled charge-discharge tests.

Research objectives

- Create a reproducible battery-cycling dataset and data card
- Reach a mean absolute state-of-health error below 3% on held-out cycles
- Demonstrate inference under 100 ms on a low-power edge device
- Document uncertainty, failure modes and safe-use limits

Skills and preparation

- Python and signal processing
- Embedded C/C++
- Time-series machine learning
- Experimental documentation

Expected deliverables

- Versioned dataset and preprocessing pipeline
- Quantized edge model and benchmark report
- Hardware demonstrator
- Reproducibility and safety brief

Indicative timeline

Weeks 1-3	literature review and protocol
Weeks 4-7	data preparation and baselines
Weeks 8-11	edge optimization
Weeks 12-14	validation and final demonstration

Responsible-use note

This brief describes a staging research concept. It does not promise funding, credit, authorship, laboratory access, safety clearance or selection. The faculty lead must approve the final protocol, data access, ethics and safety requirements before work begins.

Apply through the authenticated NSUT Connect project page. Do not send passwords, secret keys, medical information or confidential research data with an expression of interest.